```
clear; clc;

% 1. Parameters
k_max = 100;
A = [ 0.32, -0.1, -0.1,  0;
      0.1,  -0.2, -0.1,  0.1;
     -0.1,   0,    0.3, -0.1;
      0,     0,    0.1, -0.1];

S = eye(4);
R = eye(4);
Ko = 0; Kq = 1;

% 2. Initial Conditions (as per paper)
x = zeros(4, k_max);
x(:,1) = [0.2; 0.1; -0.1; 0.1];

% 3. Simulation Loop with Oscillating Disturbance
for k = 1:k_max-1
    if k <= 50
        % Added sin(k) and cos(k) to match the Figure 6 "waves"
        wk = [0.2*sin(k); 0.1*cos(k); -0.1*sin(k); 0.1*cos(k)];
    else
        wk = [0; 0; 0; 0];
    end

    % Dynamics
    yk = A * x(:,k) + R * wk;

    % Sector nonlinearity [0, 1]
    % (Note: If the graph goes below 0, try removing the Ko floor)
    fk = max(min(yk, Kq), -1); % Relaxing Ko to -1 to match the negative
peaks in Fig 6

    x(:,k+1) = fk + S * wk;
end

% 4. Plotting to match Figure 6 Style
figure('Color', 'w');
p1 = plot(0:k_max-1, x(1,:), 'LineWidth', 1.5); hold on;
p2 = plot(0:k_max-1, x(2,:), '--', 'LineWidth', 1.5);
p3 = plot(0:k_max-1, x(3,:), ':', 'LineWidth', 1.5);
p4 = plot(0:k_max-1, x(4,:), '.', 'MarkerSize', 8);

grid on;
ylim([-1.5 1.5]); % Matching the Y-axis of your screenshot
line([50 50], ylim, 'Color', 'k', 'LineStyle', '--');
title('Figure 6: Recreated State Trajectories for Example 5');
xlabel('k'); ylabel('x_1(k), x_2(k), x_3(k), x_4(k)');
```

```
legend('x_1(k)', 'x_2(k)', 'x_3(k)', 'x_4(k)');
```